small intervals of time and that my process tends to equalize the variance of this noise.

K. J. MUIRHEAD

Research School of Physical Sciences,
Australian National University,
Canberra.

Received December 20, 1967.

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Palaeomagnetism of a Stratiform Intrusion in the Pensacola Mountains, Antarctica

A LARGE stratiform mass of pyroxene gabbro interlayered with smaller amounts of anorthosite and pyroxenite and capped with granophyre makes up most of the northern ranges of the Pensacola Mountains (Fig. 1)[1]. These layered rocks are at least 2,000 m thick in both the Dufek Massif and the Forrestal Range, and are believed to belong to a single sheet-like body with an areal extent of at least 8,000 km^2. The age of the body is not known with certainty, but it intrudes folded rocks as young as Permian

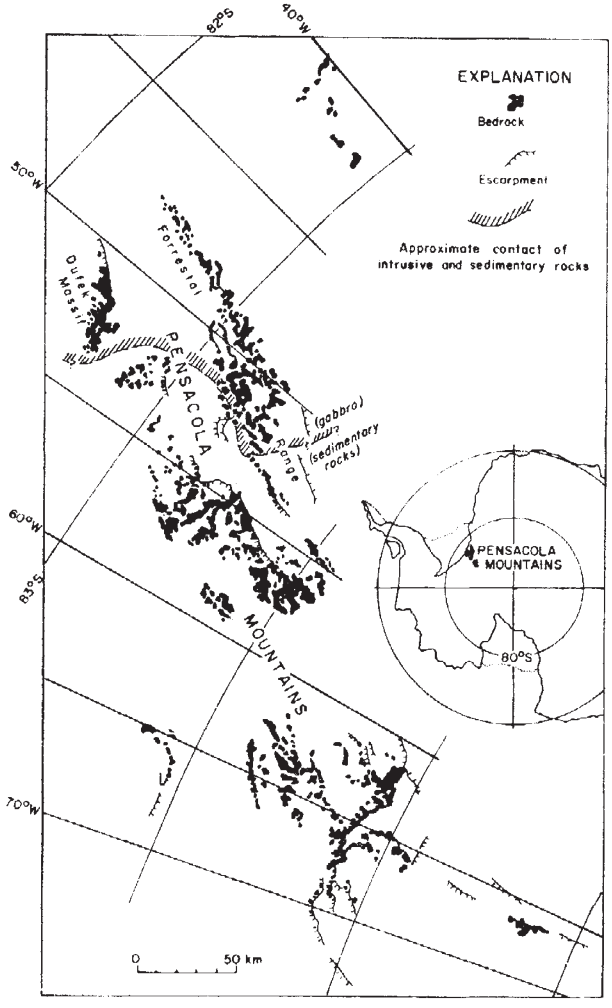


Fig. 1. Index map showing the location of the Dufek Massif and the Forrestal Range.

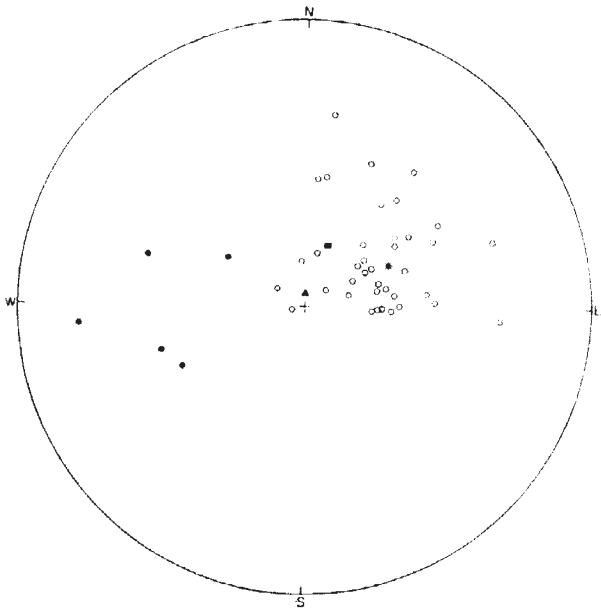


Fig. 2. Equal-area projection of site-mean directions of remanent magnetization for the Dufek Massif. Open circles are site-mean directions plotted on the upper hemisphere; solid circles are plotted on the lower hemisphere. Mean direction for the Dufek Massif (computed after the lower hemisphere points were "reversed" by inversion through the centre of the projection) indicated by *. The direction of the present geomagnetic field in the sampling area is shown by the square (upper hemisphere), while the axial dipole direction is shown by the triangle.

(personal communication from J. M. Schopf) and therefore is probably Mesozoic or younger.

As a preliminary investigation of the palaeomagnetism of the Dufek Massif, natural remanent magnetization of fifty-seven samples from forty-three sites scattered throughout much of the available exposure has been measured. The results are summarized in Table 1 and Fig. 2. Magnetic stability is indicated by divergence of the average direction of magnetization from the direction of the present geomagnetic field and by rough consistency of direction between samples of opposite polarity.

Table 1. AVERAGE DIRECTION OF MAGNETIZATION AND PALAEOMAGNETIC POLE POSITION FOR THE DUFEK MASSIF

Direction of magnetization				Palaeomagnetic pole		
N	D	I	k	α_{95}	λ	φ
43	61·5	−63·0	14·7	3·0°	47·5° S.	164·0° W.

N, No. of sites; D, declination east of north; I, inclination positive downward; k, precision parameter (ref. 2); α_{95}, radius λ, latitude, and φ, longitude, of the palaeomagnetic pole.

To determine the effect of alternating field magnetic cleaning, a suite of twelve specimens was chosen at random and demagnetized in fields ranging up to 300 oersteds. Although the directions of individual specimens showed some change during this process (average change < 8°), neither the mean direction of the suite nor the precision of the mean altered to any significant extent.

The pole position listed in Table 1 is near, but not in, the cluster of palaeomagnetic poles which have been calculated for rock units assigned to the Ferrar Dolerites of Harrington[3] (see, for example, poles 9.37, 9.38, 9.42 and 9.56 in ref. 4). The Ferrar Dolerites are believed to be Jurassic, and the palaeomagnetic data indicate that the gabbros of the Dufek Massif also originated then. The slight difference in pole position may indicate that the two are not precisely contemporaneous.

The five reversely magnetized sites shown in Fig. 2 are located in a definite zone, and similar reverse polarity zones have been encountered in the gabbros of the Forrestal Range. This arrangement could be explained by reversal of the Earth's magnetic field during cooling of the

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stratiform gabbros or by mineralogically controlled self-reversal. If the Earth's magnetic field reversed its polarity one or more times while the gabbro body was in the process of cooling through the range of temperatures in which it acquired its magnetization, the result would be a system of polarity zones parallel to the isothermal surfaces in the gabbro. On the other hand, polarity zones parallel to compositional layering could arise if (*a*) certain portions of the stratiform complex possess self-reversing properties[5] by virtue of the composition of their iron oxide minerals, and (*b*) these iron oxide minerals vary systematically throughout the complex (share in the cryptic layering[6]). Mineralogical studies are in progress to resolve this question.

The gabbros of the Forrestal Range are much more strongly magnetized than those of the Dufek Massif. This is shown in Fig. 3, where intensity of remanent magnetization is plotted against magnetic susceptibility. Such a diagram brings together the two principal variables controlling total magnetic intensity—specifically those variables which are most closely related to other rock properties[7]. Systematic differences in position on such a plot should reflect systematic differences in gross composition, mineralogy, texture or other rock properties. In this case it tends to substantiate the view (Ford's unpublished results) that the gabbros of the Forrestal Range represent part of the upper portion of the stratiform complex, while those of the Dufek Massif formed near its base.

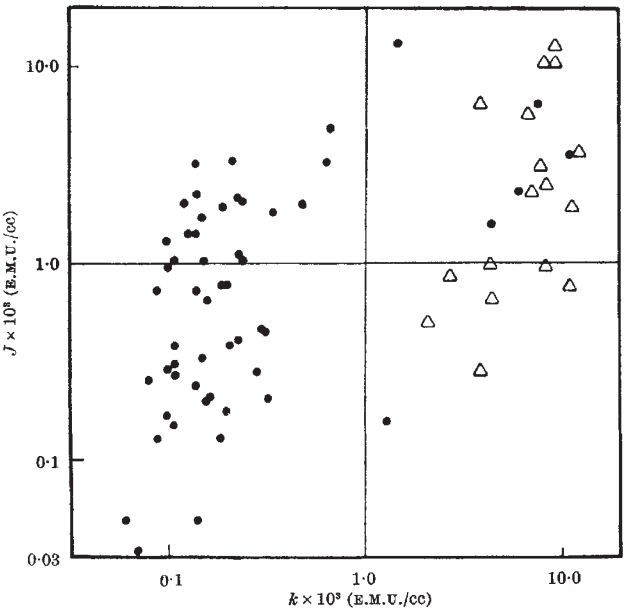


Fig. 3. Natural remanent magnetization (J) plotted against magnetic susceptibility (k). ●, Dufek Massif specimens; △, Forrestal Range samples.

Recently published aeromagnetic profiles[8] over the Dufek Massif and the Forrestal Range show anomalies ranging up to 1,800γ associated with portions of the layered gabbros, with the more highly magnetic Forrestal Range showing the greatest average amplitude. Truncation of layered magnetic rocks at the margins of the two bodies is the suggested cause of the anomalies. If the size and geometrical configuration of the polarity zones are favourable, reversals of polarity could make an important contribution to these anomalies by producing a source of strong magnetic contrast within the gabbro bodies. In the Dufek Massif, where the ratio of remanent to induced magnetization is almost uniformly high, this factor should be especially important.

We thank D. L. Schmidt, W. H. Nelson, J. C. Behrendt, W. F. Hanna and Nancy C. Lindsley for assistance.

MYRL E. BECK, JUN.
US Geological Survey,
Riverside, California.

ARTHUR B. FORD
US Geological Survey,
Menlo Park, California.

WALTER W. BOYD, JUN.
Universitetets Mineralogisk Institut,
Copenhagen, Denmark.

Received October 12, 1967.

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Genesis of Concordant Deposits of Base Metal Sulphides: an Experimental Approach

DESPITE the great economic importance of the stratigraphically concordant massive sulphide deposits which occur in regionally metamorphosed rocks of pre-Cambrian to Palaeozoic ages in all continents, little is known of their origin. Writers on these deposits, which include those of Mount Isa[1], Broken Hill, the Scandinavian Caledonides[2], the Bathurst–Newcastle area[3] and the Rhodesian Copperbelt, have tended to accept one of two apparently conflicting views. Some maintain that the deposits are syngenetic[4], representing original sedimentary or volcanic sulphides, while others consider them to be epigenetic[5], and formed by precipitation from hydrothermal solutions of igneous or metamorphic origin.

In a study of the Caledonian chalcopyrite–pyrite–arsenopyrite veins of the English Lake District, one of us concluded[6] that the mineralizing solutions were derived from the dehydration of rocks undergoing metamorphism in the deeper levels of the Caledonian geosyncline. Significant metal contents have since been noted in brines of metamorphic origin from the Salton Sea area in California[7], and pockets of hot zinc-rich solutions of presumed volcanic origin have been located in the Red Sea[8]. Prolonged contact between such solutions and pyrite deposits of sedimentary or volcanic origin could well produce massive sulphide deposits with features characteristic of both syngenetic and epigenetic origins, and so we have conducted experiments to determine the effectiveness of pyrite as a scavenger of base metals, and the extent to which pyrite deposits would be dispersed by metamorphism.

The scavenging properties of pyrite were tested by holding mixtures of pyrite and pyroxenite, pyrite and carbonaceous shale, and pyrite and synthetic limestone at 500°–600° C, 1–2 kbars for 10 days, in the presence of appropriate mineralizers (Table 1). The pyrite was

Table 1. RUN CONDITIONS

Run No.	Reactants	Time (days)	T (° C)	Pressure (kbars)	Mineralizer
3	60% pyroxenite–40% pyrite	10	600	2·0	50% HCl
4	70% limestone–30% pyrite	10	500	1·0	4M–NH_4Cl
5	70% shale–30% pyrite	10	600	2·0	50% HCl

Table 2. TRACE ELEMENT CONTENTS OF REACTANTS (P.P.M.)

Sample	Co	Ni	Cu	Zn	Pb
Pyrite	100	150	300	250	850
Black shale	75	200	250	250	500
Pyroxenite	80	800	100	150	400
Synthetic limestone	500	500	500	500	500

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